ELECTRICAL CIRCUIT AND NETWORKS

K) Suggested Laboratory (Practical) Session Outcomes (LSOs) and List of Practical: P2420301

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 1.1. Identify the commonly used components in an electrical circuit.	1.	Identification of components used in the given electrical Circuit	CO1
LSO 1.2. Measure voltage and current using suitable meters/instruments in the given linear electric circuit.	2.	Measurement of voltage and current in a given linear electrical circuit.	CO1
LSO 1.3. Measure current and voltage in a given electric circuit by applying Kirchhoff 's Current law.	3.	Measurement of current and voltage in a branch of the given electrical circuit using Kirchhoff's Current Law.	CO1
LSO 1.4. Measure voltage drop in a closed loop in a given electric circuit by applying Kirchoff's Voltage Law.	4.	Measurement voltage drop in closed loop of the given electrical circuit using Kirchhoff's Voltage Law.	CO1
LSO 1.5. Connect star connected resistances to its equivalent delta connection and determine the equivalent resistance.	5.	Connection of star connected resistances to its equivalent delta connection to measure the equivalent resistance.	CO1
LSO 1.6. Connect delta connected resistances to its equivalent Star connection and determine the equivalent resistance.	6.	Connection of delta connected resistances to its equivalent Star connection to measure the equivalent resistance.	CO1
LSO 1.7. Measure current and voltage of the given electric circuit using mesh analysis technique.	7.	Application of mesh analysis to measure current and voltage of the given electric circuit.	CO1
LSO 1.8. Measure voltage across a circuit element of a given electric circuit applying nodal analysis technique.	8.	Application of nodal analysis to measure voltage across a circuit element of a given electric circuit	CO1
LSO 2.1. Measure current in a branch of thegiven bilateral multiple source circuit using superposition theorem.	9.	Measurement of current in a branch of the given electrical circuit having two or more input sources using Super position theorem.	CO1, CO2
LSO 2.2. Determine the circuit parameters ofthe given network using Thevenin's theorem.	10.	Measurement of load current in the load resistance using Thevenin's theorem in a given circuit.	CO1, CO2
LSO 2.3. Determine the circuit parameters ofthe given network using Norton's theorem.	11.	Measurement of load current in the load resistance using Norton's theorem in a given circuit.	CO1, CO2
LSO 2.4. Measure the value of load resistance for which maximum power is produced in the given electric circuit.	12.	Measurement of the value of load resistance for which maximum power is produced in a given electric circuit.	CO1, CO2
LSO 3.1. Measure the peak value, RMS value, Period and frequency of a sinusoidal voltage using CRO.	13.	Measurement of peak value, RMS value, Period and frequency of a sinusoidal voltage using CRO.	CO2, C03
LSO 3.2. Plot the waveform of voltage and current in a resistive load using CRO.	14.	Waveform of voltage and current in a resistive load.	CO2, C03
LSO 3.3. Plot the waveform of voltage and current in an R-L load.	15.	Plot the waveform of voltage and current in a R-L load.	CO2, C03
LSO 3.4. Plot the waveform of voltage and current in an R-L-C load.	16.	Plot the waveform of voltage and current in a R-L-C load	CO2, C03
LSO 3.5. Measure the voltage, current in a series RLC circuit and calculate power and power factor and draw phasor diagram.	17.	Measurement of voltage, current, power and power factor in a series RLC circuit	CO2, C03

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 3.6. Measure voltage, current, power and power factor in an RLC parallel circuit and draw phasor diagram	18.	Measurement of voltage, current, power and power factor in a RLC parallel circuit	CO2, C03
LSO 3.7. Determine the power and power factor in AC circuit using three ammeter method.	19.	Determination of the power and power factor in AC circuit using three ammeter method	CO2, C03
LSO 4.1. Determine the current at resonance in a series RLC circuit	20.	Determination of the current in an electric circuit at series resonance.	C03, CO4
LSO 4.2. Determine the resonance frequency and impedance of the given parallel RLC circuit at resonance	21.	Determination of the resonance frequency and impedance of the given parallel RLC circuit at resonance	C03, CO4
LSO 4.3. Measure Open Circuit (Z) parameter, Short Circuit (Y) of a two-port network	22.	Measurement of Open Circuit (Z) parameter, Short Circuit (Y) of a two-port network	C03, CO4
LSO 5.1. Measure the he line/phase current, line voltage/phase voltage for the given three phase loads connected to a three-phase source.	23.	Measurement of the line/phase current, line voltage/phase voltage for the given three phase loads connected to a three-phase source.	CO5
LSO 5.2. Measure three phase power for the given star connected load.	24.	Measurement of neutral displacement voltage of the given three phase unbalanced load connected to a three-phase source	CO4, CO5
LSOs 5.3 Measure three phase power for the given star/delta connected load	25.	Measurement of three phase power for the given star/delta connected load	CO4, CO5

L) Suggested Term Work and Self Learning: S2420301 Some sample suggested assignments, micro project and other activities are mentioned here for reference.

a. **Assignments**: Questions/Problems/Numerical/Exercises to be provided by the course teacher in line with the targeted COs.

Unit-5.0 Three Phase AC circuits	12	CO4, CO5	16	5	5	6
Total	**48**	**-**	**70**	**20**	**25**	**25**

Note: Similar table can also be used to design class/mid-term/ internal question paper for progressive assessment.

O) Suggested Assessment Table for Laboratory (Practical):

S. No.	Laboratory Practical Titles	Relevant COs Number(s)	PLA/ELA		
			Performance		Viva-Voce
			PRA* (%)	PDA** (%)	(%)
1.	Identification of components used in the given electrical Circuit	CO1	50	40	10
2.	Measurement of voltage and current in a given linear electrical circuit.	CO1	60	30	10
3.	Measurement of current and voltage in a branch of the given electrical circuit using Kirchhoff's Current Law.	CO1	60	30	10
4.	Measurement voltage drop in closed loop of the given electrical circuit using Kirchhoff's Voltage Law.	CO1	60	30	10
5.	Connection of star connected resistances to its equivalent delta connection to measure the equivalent resistance.	CO1	50	40	10
6.	Connection of delta connected resistances to its equivalent Star connection to measure the equivalent resistance.	CO1	50	40	10
7.	Application of mesh analysis to measure current and voltage of the given electric circuit.	CO1	50	40	10
8.	Application of nodal analysis to measure voltage across a circuit element of a given electric circuit	CO1, CO2	50	40	10
9.	Measurement of current in a branch of the given electrical circuit having two or more input sources using Super position theorem.	CO1, CO2	50	40	10
10.	Measurement of load current in the load resistance using Thevenin's theorem in a given circuit.	CO1, CO2	40	50	10
11.	Measurement of load current in the load resistance using Norton's theorem in a given circuit.	CO1, CO2	40	50	10

S. No.	Laboratory Practical Titles	Relevant COs Number(s)	PLA/ELA		
			Performance		Viva-Voce
			PRA* (%)	PDA** (%)	(%)
12.	Measurement of the value of load resistance for which maximum power is produced in a given electric circuit.	CO1, CO2	60	30	10
13.	Measurement of peak value, RMS value, Period and frequency of a sinusoidal voltage using CRO.	CO1, CO3	50	40	10
14.	Plot the waveform of voltage and current in a resistive load using CRO.	CO1, CO3	50	40	10
15.	Plot the waveform of voltage and current in a R-L load.	CO1, CO3	50	40	10
16.	Plot the waveform of voltage and current in a R-L-C load	CO1, CO3	50	40	10
17.	Measurement of voltage, current, power and power factor in a series RLC circuit	CO1, CO3	50	40	10
18.	Measurement of voltage, current, power and power factor in a RLC parallel circuit	CO1, CO3	45	45	10
19.	Determination of the power and power factor in AC circuit using three ammeter method	CO1, CO3	50	40	10
20.	Determination of the current in an electric circuit at series resonance.	CO3, CO4	50	40	10
21.	Determination of the resonance frequency and impedance of the given parallel RLC circuit at resonance	CO3, CO4	50	40	10
22.	Measurement of Open Circuit (Z) parameter, Short Circuit (Y) of a two-port network	CO4	50	40	10
23.	Measurement of the line/phase current, line voltage/phase voltage for the given three phase loads connected to a three-phase source.	CO1, CO3, CO5	60	30	10
24.	Measurement of neutral displacement voltage of the given three phase unbalanced load connected to a three-phase source	CO5	50	40	10
25.	Measurement of three phase power for the given star/delta connected load	CO5	50	40	10

Legend:
PRA*: Process Assessment
PDA**: Product Assessment

Note: This table can be used for both end semester as well as progressive assessment of practical. Rubrics need to be prepared by the course teacher for each experiment/practical to assess the student performance.

ELECTRICAL MEASUREMENTS AND INSTRUMENTATION

K) Suggested Laboratory (Practical) Session Outcomes (LSOs) and List of Practical: P2420302

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 1.1 Select Indicating, Recording and Integrating Instruments in your laboratory and write their specifications and features.	1.	Identification of Indicating, Recording and Integrating Instruments	CO1
LSO 2.1 Measure DC, AC voltage and current using analogue meter.	2.	Measurement of DC, AC voltage and current.	CO2
LSO 2.2 Convert a given galvanometer to DC/AC current- meter.	3.	Conversion of a given galvanometer to DC/AC current- meter.	CO2
LSO 2.3 Extend the range of ammeter and voltmeter to measure high value of current and voltages using shunt and multiplier.	4.	Extension of range of ammeter and voltmeter to measure high value of current and voltages using shunt and multiplier.	CO2
LSO 2.4 Measure high value of current and voltages using Current and Potential Transformer.	5.	Measurement of high value of current and voltages using Current and Potential Transformer.	CO2
LSO 2.5 Calibrate the given ammeter and voltmeter with a standard meter	6.	Calibration of ammeter and voltmeter	CO2
LSO 2.6 Interpret the working principle of moving iron and moving coil type instruments.	7.	Demonstration of working of Moving Iron and Moving Coil type instruments.	CO2
LSO 2.7 Interpret the working principle of Induction type and dynamometer type instruments	8.	Demonstration of the working of Induction type and dynamometer type instruments	CO2
LSO 2.8 Measure voltage, current, resistance using Digital Multimeter	9.	Measurement of voltage, current, resistance using Digital Multimeter	CO2
LSO 2.9 Perform continuity test using digital Multimeter	10.	Continuity test using digital Multimeter	CO2
LSO 3.1 Measure single and three phase power using one wattmeter	11.	Measurement of single and three phase power using one wattmeter	CO3
LSO 3.2 Measure 3 phase power using two and three wattmeter method	12.	Measurement of 3 phase power using two and three wattmeter method	CO3
LSO 3.3 Calibrate the given wattmeter with a standard meter.	13.	Calibration of wattmeter.	CO3
LSO 3.4 Calibrate the given single-phase energy meter with a standard meter.	14.	Calibration of single-phase energy meter.	CO3
LSO 3.5 Interpret the working of a digital energy meter	15.	Demonstration of the working of a digital energy meter	CO3
LSO 4.1 Use Kelvin's double bridge for measurement of low resistance	16.	Kelvin's double bridge for measurement of low resistance	CO4
LSO 4.2 Measure medium resistance using Wheatstone bridge or Voltmeter-Ammeter method or Ohmmeter	17.	Measurement of medium resistance using Wheatstone bridge or Voltmeter-Ammeter method or Ohmmeter	CO4
LSO 4.3 Us Megger to measure insulation resistance. *LSO 4.4* Measure insulation resistance using Megger.	18.	Measurement of insulation resistance using Megger.	CO4
LSO 4.5 Using Earth tester to measure earth resistance. *LSO 4.6* Measure earth resistance using Earth tester.	19.	Measurement of earth resistance using Earth tester	CO4

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 4.7 Use Anderson or Maxwell inductance Capacitance Bridge to measure unknown inductance. *LSO 4.8* Measure unknown inductance using Anderson or Maxwell inductance capacitance bridge	20.	Measurement of unknown inductance using Anderson or Maxwell inductance capacitance bridge	CO4
LSO 4.9 Use Schering's Bridge to measure unknown capacitance. *LSO 4.10* Measure unknown capacitance using Schering's Bridge.	21.	Measurement of unknown capacitance using Schering's Bridge	CO4
LSO 4.11 Use Wein bridge or Weston frequency meter to measure unknown frequency *LSO 4.12* Measure unknown frequency using Wein Bridge or Weston frequency meter.	22.	Measurement of unknown frequency using Wein bridge or Weston frequency meter	CO4
LSO 5.1 Measure power factor using a dynamometer PF meter	23.	Measurement of power factor using a dynamometer PF meter	CO5
LSO 5.2 Use phase sequence indicator to identify the phase sequence and reverse the phase sequence.	24.	Identification of phase sequence and reverse the phase sequence using the phase sequence indicator	CO5
LSO 5.3 Demonstrate the use of Synchroscope for Synchronization	25.	Demonstration of use of Synchroscope for Synchronization	CO5
LSO 5.4 Measure the amplitude, frequency, time period and Phase difference of different signals generated by function generator using CRO.	26.	Measurement of amplitude, Frequency, time period and Phase difference of different signals generated by function generator using CRO.	CO5
LSO 5.5 Measure Unknown frequency, phase angle using Lissajous patterns.	27.	Measurement of Unknown frequency, phase angle using Lissajous patterns.	CO5
LSO 5.6 Identify the various parts of digital storage oscilloscope.	28.	Demonstration of features of digital storage oscilloscope.	CO5
LSO 5.7 We LCR meter to measure resistance, Inductance and Capacitance. LSO 5.8 Measure resistance, Inductance and Capacitance using LCR meter.	29.	Measurement of resistance, Inductance and Capacitance using LCR meter	CO5
LSO 5.9 Measure quality Factor of the given Inductor and Capacitor using LCR/Q Meter	30.	Measurement of quality Factor of Inductor and Capacitor using LCR/Q Meter	CO5
LSO 5.10 Interpret the working principle of various analog/digital recorders.	31.	Demonstration of the working of various analog/digital recorders.	CO5

L) **Suggested Term Work and Self Learning: S2420302** Some sample suggested assignments, micro

O) Suggested Assessment Table for Laboratory (Practical):

S. No.	Laboratory Practical Titles	Relevant Cos Number (s)	PLA/ELA Performance PRA* (%)	PDA** (%)	Viva-Voce (%)
1.	Identification of Indicating, Recording and Integrating Instruments	CO1	50	40	10
2.	Measurement of DC, AC voltage and current.	CO2	50	40	10
3.	Conversion of a given galvanometer to DC/AC current- meter.	CO2	50	40	10
4.	Extension of range of ammeter and voltmeter to measure high value of current and voltages using shunt and multiplier.	CO2	50	40	10
5.	Measurement of high value of current and voltages using Current and Potential Transformer.	CO2	50	40	10
6.	Calibration of ammeter and voltmeter	CO2	50	40	10
7.	Demonstration of working of Moving Iron and Moving Coil type instruments.	CO2	50	40	10
8.	Demonstration of the working of Induction type and dynamometer type instruments	CO2	50	40	10
9.	Measurement of voltage, current, resistance using Digital Multimeter	CO2	50	40	10
10.	Continuity test using digital Multimeter	CO2	50	40	10
11.	Measurement of single and three phase power using one wattmeter	CO3	50	40	10
12.	Measurement of 3 phase power using two and three wattmeter method	CO3	50	40	10
13.	Calibration of wattmeter.	CO3	50	40	10
14.	Calibration of single-phase energy meter.	CO3	50	40	10
15.	Demonstration of the working of a digital energy meter	CO3	50	40	10
16.	Kelvin's double bridge for measurement of low resistance	CO4	50	40	10
17.	Measurement of medium resistance using Wheatstone bridge or Voltmeter-Ammeter method or Ohmmeter	CO4	50	40	10
18.	Measurement of insulation resistance using Megger.	CO4	50	40	10
19.	Measurement of earth resistance using Earth tester	CO4	50	40	10
20.	Measurement of unknown inductance using Anderson or Maxwell inductance capacitance bridge	CO4	50	40	10
21.	Measurement of unknown capacitance using Schering's Bridge	CO4	50	40	10
22.	Measurement of unknown frequency using Wein bridge or Weston frequency meter	CO4	50	40	10
23.	Measurement of power factor using a dynamometer PF meter	CO5	50	40	10

S. No.	Laboratory Practical Titles	Relevant Cos Number (s)	PLA/ELA Performance PRA* (%)	PDA** (%)	Viva-Voce (%)
24.	Identification of phase sequence and reverse the phase sequence using the phase sequence indicator	CO5	50	40	10
25.	Demonstration of use of Synchroscope for Synchronization	CO5	50	40	10
26.	Measurement of amplitude, Frequency, time period and Phase difference of different signals generated by function generator using CRO.	CO5	50	40	10
27.	Measurement of Unknown frequency, phase angle using Lissajous patterns.	CO5	50	40	10
28.	Demonstration of features of digital storage oscilloscope.	CO5	50	40	10
29.	Measurement of resistance, Inductance and Capacitance using LCR meter	CO5	50	40	10
30.	Measurement of quality Factor of Inductor and Capacitor using LCR/Q Meter	CO5	50	40	10
31.	Demonstration of the working of various analog/digital recorders.	CO5	50	40	10

Legend:
PRA*: Process Assessment
PDA**: Product Assessment

Note: This table can be used for both end semester as well as progressive assessment of practical. Rubrics need to be prepared by the course teacher for each experiment/practical to assess the student performance.

P) Suggested Instructional/Implementation Strategies: Different Instructional/ Implementation

	5.5 Potential Transformer 5.6 Welding transformer	

Note: One major TSO may require more than one theory session/period.

K) Suggested Laboratory (Practical) Session Outcomes (LSOs) and List of Practical: P2420303

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 1.1 Identify different parts of a DC machine.	1.	Identification of parts of a DC Machine by dismantling the cut section model of a DC machine	CO1
LSO 1.2 Interpret the effect of speed and field flux on generated voltage of DC shunt generator.	2.	Effect of speed and field flux on generated voltage of DC shunt generator	
LSO 1.3 Test the performance of DC shunt generator on the given load condition.	3.	Load test of DC shunt Generator	
LSO 1.4 Test the performance of DC series generator on the given load condition.	4.	Load test of DC series Generator	
LSO 2.1 Use appropriate DC motor starter for starting the given DC Motor.	5.	Starting of D. C shunt motor using 3-point /4-point starter	CO1, CO2
LSO 2.2 Change e terminal connection of DC shunt motor and observe the direction of rotation	6.	Reversal of Direction of a DC Shunt motor	
LSO 2.3 Control speed of DC Shunt motor using field/flux control method (Above rated speed)	7.	Speed control of D.C shunt motor	
LSO 2.4 Control speed of DC Shunt motor using armature control method (Below rated speed).	8.	Speed control of a D.C. Shunt motor	

Practical/Lab Session Outcomes (LSOs)	S. No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 2.5 Test performance of DC shunt motor on the given load condition.	9.	Load test of D. C. shunt motor	
LSO 2.6 Test performance of DC series motor on the given load condition.	10.	Load test of D.C. series motor	
LSO 2.7 Apply direct method to brake the DC shunt motor.	11.	Brake test of D.C. shunt motor.	
LSO 3.1 Measure voltage and current ratio of a given single phase transformer	12.	Measurement of voltage and current ratio of a given single phase transformer.	CO3
LSO 3.2 Test polarity of the given single phase transformer.	13.	Polarity of a single-phase transformer	
LSO 3.3 Test performance of a given transformer by direct load test.	14.	Direct load test on a single-phase transformer	
LSO 3.4 Observe the no load waveform of a given transformer using CRO	15.	No load waveform of a transformer using CRO	
LSO 3.5 Perform Open Circuit and Short Circuit test on a single-phase transformer.	16.	Open Circuit and Short Circuit test on a single-phase transformer.	
LSO 3.6 Perform parallel operation of two 1-phase transformer having equal and unequal KVA rating under given load.	17.	Parallel operation of two single phase transformers.	
LSO 3.7 Test performance of an auto transformer and 1-phase two winding transformer of same rating.	18.	Auto transformer and 1-phase two winding transformer of same rating	
LSO 4.1 Perform parallel operation of two 3-phase transformer having equal and unequal load sharing for a given load.	19.	Parallel operation of two three phase transformers for equal and unequal load sharing.	CO3, CO4
LSO 5.1 Measure current ratio of the given Current transformer using ammeter/clamp meter.	20.	Current ratio of a Current transformer	CO4, CO5
LSO 5.2 Measure voltage ratio of the given Potential transformer using voltmeter.	21.	Voltage ratio of a Potential Transformer	CO4, CO5

O) Suggested Assessment Table for Laboratory (Practical):

S. No.	Laboratory Practical Titles	Relevant Cos Number (s)	PLA /ELA		
			Performance		Viva-Voce (%)
			PRA* (%)	PDA** (%)	
1.	Identification of parts of a DC Machine by dismantling the cut section model of a DC machine	CO1	50	40	10
2.	Effect of speed and field flux on generated voltage of DC shunt generator	CO1	50	40	10
3.	Load test of DC shunt Generator	CO1	50	40	10
4.	Load test of DC series Generator	CO1	50	40	10
5.	Starting of D. C shunt motor using 3-point /4-point starter	CO2	50	40	10
6.	Reversal of direction of a DC Shunt motor	CO2	50	40	10
7.	Speed control of D.C shunt motor	CO2	50	40	10
8.	Speed control of a DC Shunt motor	CO2	50	40	10
9.	Load test of D. C shunt motor	CO2	50	40	10
10.	Load test of D.C. series motor	CO2	50	40	10
11.	Brake test of D.C. shunt motor.	CO2	50	40	10
12.	Measurement of voltage and current ratio of a given single phase transformer.	CO3	50	40	10
13.	Polarity of a single-phase transformer	CO3	50	40	10
14.	Direct load test on a single-phase transformer	CO3	50	40	10
15.	No load waveform of a transformer using CRO	CO3	50	40	10
16.	Open Circuit and Short Circuit test on a single-phase transformer.	CO3	50	40	10
17.	Perform Parallel operation of two single phase transformers.	CO3	50	40	10
18.	Auto transformer and 1-phase two winding transformer of same rating	CO3	50	40	10
19.	Parallel operation of two three phase transformers for equal and unequal load sharing	CO4	50	40	10
20.	Current ratio of a Current transformer	CO5	50	40	10
21.	Voltage ratio of a Potential Transformer	CO5	50	40	10

	text files, reading and writing binary files, the access modes	

Note: One major TSO may require more than one Theory session/Period.

K) Suggested Laboratory (Practical) Session Outcomes (LSOs) and List of Practical: P2418305

Practical/Lab Session Outcomes (LSOs)	S.No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 1.1. Write, execute and debug simple Python program using Integrated Development and Learning Environment (IDLE) *LSO 1.2.* Write and execute simple 'C' program using variables, arithmetic expressions.	1.	a) Download and Install IDLE. Write and execute Python program to- b) Calculate the Area of a Triangle where its three sides a, b, c are given. s=(a+b+c)/2, Area=square root of s(s-a)(s-b)(s-c) (write program without using function) c) Swap Two Variables d) Solve quadratic equation for real numbers.	CO-1
LSO 2.1. Write and execute python programs using conditional statements. *LSO 2.2.* Write and execute python programs using various types of Loop statements	2.	Write and execute Python program to- a) Check if a Number is Positive, Negative or zero. b) Check whether the given year is a Leap Year. c) Print all Prime Numbers in an Interval. d) Display the multiplication Table based on the given input. e) Print the Fibonacci sequence. f) Find the Factorial of a Number.	CO-2

Practical/Lab Session Outcomes (LSOs)	S.No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
LSO 3.1. Write and execute Python program to perform various operations on string using string operators and methods	3.	Write and execute Python program to- a) Check whether the string is Palindrome b) Reverse words in a given String in Python c) Identify in a strings the name, position and counting of vowels. d) Count the Number of matching characters in a pair of string (set) e) Python program for removing i-th character from a string	CO-2, CO-3
LSO 4.1. Write and execute Python program to perform various operations on List using List operators and methods	4.	Write and execute Python program to- a) Find largest number in a given list without using max (). b) Find the common numbers from two lists. c) Create a list of even numbers and another list of odd numbers from a given list. d) To find number of occurrences of given number without using built-in methods.	CO-2, CO-3
LSO 5.1. Write and execute Python program to perform various operations on Tuple using Tuple operators and methods.	5.	Write and execute Python program to- a) Find the index of an item of a tuple. b) Find the length of a tuple. c) To reverse a tuple. d) Write a Python program to sort a list of tuple by its float element. Sample data: [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')] Expected Output: [('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]	CO-2, CO-3
LSO 6.1. Write and execute Python program to perform various operations on sets using set methods.	6.	Write and execute Python program to- a) Create an intersection of sets. b) Create a union of sets. c) Create set difference. d) Check if two given sets have no elements in common.	CO-2, CO-3
LSO 7.1. Write and execute Python program to perform various operations on Dictionary using Dictionary methods	7.	Write and execute Python program to- a) Write a Python script to concatenate two dictionaries to create a new one b) Write a Python script to merge two Python dictionaries. c) Write a Python program to combine two dictionary adding values for common keys. d1 = {'a': 100, 'b': 200, 'c':300} d2 = {'a': 300, 'b': 200, 'd':400} Sample output: d({'a': 400, 'b': 400, 'd': 400, 'c': 300})	CO-2, CO-3
LSO 8.1. Write and execute Python program to create user defined functions and call them.	8.	Write and execute Python program to- a) Write a Python function for reversing a string and call it. b) Write a Python function for calculating compound interest and call it.	CO-2, CO-4

Practical/Lab Session Outcomes (LSOs)	S.No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
		c) Write a Python function for calculating	

Practical/Lab Session Outcomes (LSOs)	S.No.	Laboratory Experiment/Practical Titles	Relevant COs Number(s)
		c) Write a Python function for calculating the factorial of a number and call it to calculate !n/(!r)*!(n-r)) where symbol "! " stands for factorial.	
LSO 9.1. Write and execute Python program to define a numpy array. *LSO 9.2.* Develop and execute Python program Using various types of Numpy operation.	9.	a) Write a python program to create a Numpy array filled with all zeros b) Write a python program to check whether a Numpy array contains a specified row c) Write a python program to Remove rows in Numpy array that contains non-numeric values d) Write a python program to Find the number of occurrences of a sequence in a NumPy array e) Write a python program to Find the most frequent value in a NumPy array f) Write a python program to Combine a one and a two-dimensional NumPy Array g) Write a python program to Flatten a Matrix in Python using NumPy h) Write a python program to Interchange two axes of an array	**CO-2, CO-5**
LSO 10.1. Develop and execute Python program to handle various type of exceptions. *LSO 10.2.* Develop and execute Python program to perform file operations.	10.	a) Using exception handling feature such as try...except, try finally- write minimum three programs to handle following types of exceptions. i. Type Error ii. Name Error iii. Index Error iv. Key Error v. Value Error vi. IO Error vii. Zero Division Error b) Write Python program to demonstrate file operations.	**CO-6, CO-1, CO-2,**

Note: in addition to above listed practical, students are suggested to practice all the examples covered by the teacher during